

Aspect Category Detection on Indonesian E-commerce Mobile Application Review

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Abstract—E-commerce platform has a great influence on the growth of digital economy in Indonesia. This promising sector creates a fierce competition between e-commerce platforms. User reviews can be utilized to discover useful information for both developers and users. Developers use them for enhancing their application, while users take them as a consideration for using the application. We perform aspect category detection to retrieve the important aspects in reviews. We gather and analyze any aspect categories related to e-commerce and find that new potential aspects can be obtained from mobile application reviews, such as promos and payment. For identifying the aspects, we employ two different approaches, one-vs-all and single model, for 3748 annotated reviews. In one-vs-all we compare Naive Bayes, SVM, and the effect on using class-weights in SVM, since the distribution in the dataset is imbalanced. Meanwhile, we implement neural networks architecture for single model. We compare CNN to GRU+CNN architecture. GRU+CNN successfully achieves overall best result on both example-based and label-based performance metrics for multi-label task in our dataset. In example-based we use Hamming score to calculate the accuracy where GRU+CNN gets 71%, and it obtains 78% samples average F1-score for label-based metric.

Index Terms—e-commerce, application review, aspect detection, multi-label, Indonesian review

I. INTRODUCTION

The advancement of e-commerce has a great influence on the growth of digital economy in Indonesia. It becomes more appealing as it also prospers other industries, such as logistics and digital payment. The rise of smartphone ownership and ease of internet access are among factors driving the rapid development on e-commerce, as stated by McKinsey¹, around 75 percent of customers use their smartphones for online shopping. This promising sector creates a fierce competition between e-commerce platforms. Finding out their own strengths and weaknesses can be helpful to maintain and increase the quality of e-commerce platforms.

Practically, user reviews are utilized to discover those information. People often read them to obtain opinion or recommendation from other users experiences so that they can consider their decision before installing or proceeding the transactions [1]. Meanwhile, the software developers use the information for analyzing and identifying users preferences

and problems report to plan their upcoming releases [2]. Application marketplace, such as Google Play Store and Apple App Store, facilitate users to evaluate the applications in the form of rating and text review, which the later give more information, since users can assess on a specific aspect [3]. This feature is proven to be informative and helpful [3]. However, extracting information in application reviews is challenging. The amount of user reviews for each application is very large [2] yet they need to be sorted into diverse categories. They also belong to an unstructured text [4] with a lot of misspelling and informal words.

Many studies had discussed opinion mining approaches for solving various tasks. Ekanata and Budi [4], [5] conducted opinion mining to classify user reviews into categories that help them organizing requests and problem reports for maintenance or upcoming releases. Meanwhile, in [2], [3] they proposed opinion mining for summarization model. Gu and Kim [2] extracted some features or aspects from frequent words within the text using topic modelling followed by analyzing the sentiment which correlated with those features. Guzman and Maalej [3] added the app maintenance classification into the feature extraction and sentiment analysis for summarization process. The process, such as feature or aspect extraction and sentiment analysis, are also included in Aspect Based Sentiment Analysis task. Specifically, ABSA is divided into four sub-tasks i.e. aspect term extraction, aspect term polarity, aspect category detection, and aspect category polarity.

This paper focus on implementing aspect category detection for e-commerce mobile application reviews in Bahasa Indonesia, where the discussion on this domain is still limited. Aspect category detection is important to identify certain aspects which are discussed in the reviews and is useful for both application users and developers. Aspect category detection is a multi-label classification problem. The most common approach is using one-vs-all classifier. This approach with Support Vector Machine (SVM) classifier achieved one of the best result in SemEval 2014 on restaurant reviews [6]. Furthermore, one-vs-all approach was also implemented with deep neural network in SemEval 2016 using Convolutional Neural Network (CNN) layer [7], while [8] compared datasets between SemEval 2014, 2015, and 2016 using the combination of Long short-term memory (LSTM) and CNN layers. For

¹<https://theaseanpost.com/article/rise-e-commerce-indonesia>

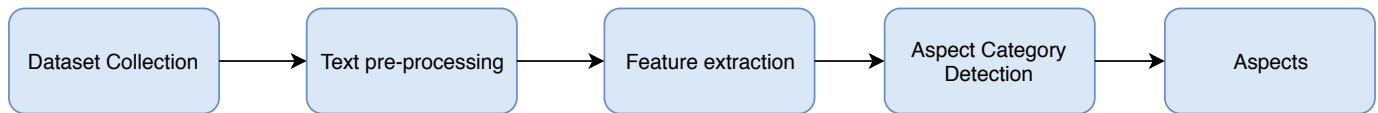


Fig. 1. Research methodology

e-commerce domain, the recent studies on detecting aspect categories was conducted using a single model of Recurrent Neural Network (RNN) [9]. Unlike one-vs-all approach, single model has the advantage that it does not need to train multiple labels independently. We perform experiments to compare the performance between one-vs-all and single model approaches on our dataset. We use classifier models such as SVM, Naive Bayes, and neural network. We also analyze more valuable aspects which haven't been discussed in previous studies.

The rest of this paper are structured as follows: Section II we describe the related works in aspect category detection for user reviews; Section III we discuss on our proposed methods; and the last, Section IV, for presenting and analyzing our results.

II. RELATED WORK

Aspect category detection is a task to identify one or more aspects in a review sentence. Each domain may have different aspects and terms to define them, for example in restaurant domain, reviews which contain terms like 'staffs service' or 'waitress' can be categorized into *service* category while in e-commerce we find 'customer service' or 'sellers response' instead. In fact, the remaining categories for restaurant and e-commerce are very different. Most similar works on detecting aspect categories in e-commerce reviews are [1], [9], [10]. Sari et al. [10] collected the dataset from an open community review website, Trustedcompany, to retrieve user reviews on one of local e-commerce platform. They focused on assessing the service quality, consisted of five aspects: *web design*, *reliability*, *responsiveness*, *trust*, and *personalization*. They applied term frequencyinverse document frequency (TF-IDF) for feature extraction with Naive Bayes classifier to predict the aspect categories since it accommodates small and balanced datasets. The classifier is said to give a high accuracy on classifying both aspects and sentiments.

Fachrina and Widyantoro [1] collected sellers' reviews, directly from the same local e-commerce platform, on several categories i.e. *electronic*, *fashion*, and *automotive*. They gathered a questionnaire to select the most important aspects and concluded with *good accuracy*, *good quality*, *service quality*, *communication*, *service rate*, *packaging quality*, *delivery time*, *price*, and *good availability*. They compared rule-based approach with machine learning algorithms, SVM and Naive Bayes, to directly classify aspect category polarity. Ilmania [9] then performed aspect category detection and aspect category polarity tasks separately. For aspect detection model, they employed RNN with gated recurrent unit (GRU) using word embedding vector, word2vec, as the input compare to fully-connected layer with bag-of-words vector. The GRU layer

contributed more for a better performance which receives 88.55% in F1-score.

For other domains, [6] achieved a good F1-score of 88.58% using one-vs-all SVM classifier in SemEval-2014 restaurant dataset with 5 aspect categories. In SemEval-2016, the aspects are more varied. The restaurant datasets consists of 12 categories for 2676 review sentences. Toh [7] ran one-vs-all approach using CNN classifier and get 75.10% in F1-score, while in [8] they combined BiLSTM layer with CNN layer to receive a better F1-score of 75.69%.

III. RESEARCH METHODOLOGY

Our research process starts with collecting the dataset as shown in Fig. 1. We focused on analysing Indonesian text reviews on various e-commerce mobile application. Mobile application reviews contain reliable information for both users and developers. We gathered many sources, including surveys and articles, to select the aspect categories. In an article², it was said that discounts and promos were among the most favorite services in e-commerce platform. Survey reports from DailySocial³ also showed the positive result on promos and discounts. Moreover, along with the growth of digital payment and cash-on-delivery (COD) facilities in Indonesia, we noticed that payment development is important to e-commerce. However these aspects had not been discussed in previous studies. We found both promos and payment aspects can be discovered in mobile marketplace reviews. We describe more details about our selected aspects in Table 1.

Afterwards, the text reviews are submitted into text pre-processing steps, consisted of *non-ASCII character removal*, *tokenization*, *text cleaning*, *stopwords removal*, and *stemming*. These steps help the text reviews to be easier for analysis. The cleaned sentences are then extracted as features for the input of classifier models.

A. Text Pre-processing

- *Non-ASCII character removal*. The text reviews include some unrecognized characters. Since the characters are insignificant to our experiments and bear no meaning, we decided to remove them. We use regex to find those non-ASCII characters and replace them with whitespace.
- *Tokenization*. In tokenization process, we employ sPacy⁴ library for Indonesian corpus to split the words. We also

²<https://www.indonesia-investments.com/id/news/todays-headlines/e-commerce-in-indonesia-many-consumers-look-offline-buy-online>

³<https://dailysocial.id/post/mengungkap-layanan-e-commerce-terpopuler-di-indonesia>

⁴<https://spacy.io/usage/models>

TABLE I
ASPECTS DESCRIPTION

Aspect	Description
app#quality	The quality of application system, e.g. app response, ease of use, user interface, user experience, bugs, or notification
delivery#cost	Cost for delivering products, including free delivery
delivery#service	Delivery processes and courier services
event#promo	Promos offering, e.g. voucher, discount, or deals
event#general	Loyalty program, e.g. games, quiz, or rewards (coin or point)
payment#option	Payment methods including bank transfer, cash on delivery, counter payment, or proprietary payment method
payment#process	Payment processes, including order payment and top up balance, payment status, or payment confirmation
product#availability	Product selection or availability of product when order
product#price	The price of products
product#quality	The quality of received products e.g. authenticity or similarities to product description
service#funds	The process of refund or withdrawal
service#general	E-commerce services and policies e.g. customer services, useful features, or user policies
service#seller	Sellers response and services e.g. difficult to contact, packing, or wrong delivered items

transform all characters into lower-case, remove non-alphabet character, and ensure the split text are not in url or email format to avoid additional unnecessary tokens.

- *Text cleaning.* The text cleaning is important to normalize the misspelling, slang, and abbreviated tokens into standard words, since those are common in online reviews. We manually created a domain-specific dictionary which contains 2594 key-value pairs.
- *Stopwords removal.* We also use our own created stopwords list for this step. The list consists of words such as conjunction words (e.g. "dan" means and, "atau" means or), Indonesian particle (e.g. "ya", "sih", and "deh") or emoticon-translated words (e.g. "hahaha" expressing laugh or "cuih" expressing disgust).
- *Stemming.* Finally, we stem the cleaned words into their roots by utilizing Sastrawi⁵ library. Stemming process help increasing the performance on information retrieval [11].

B. Aspect Category Detection Model

There are several strategies to solve multi-label text classification problem, such as aspect category detection. According to previous studies one-vs-all approach gave strong performance result, but recently experiments on single model with neural network also shows improvement.

One-vs-all transforms multi-label classification into independent binary classification for each label. We employ SVM

and Naive Bayes models to classify the single binary, 0 or 1, which indicates an aspect. The combined predicted labels from multiple classification models are taken as the result of multi-label task. Since we train classifier model individually, this approach doesn't consider any relations between aspects. Moreover, the aspects which are initially smaller, compare to other categories, are prone to imbalanced problem as this approach puts current processed aspect against others. We apply cost-sensitive learning method, class weighting, to deal with this problem. Class weighting is also considered as sampling method where minority labels are given higher weights than the majority labels, high weights imply duplication in sampling [12]. We combine TF-IDF, word unigram, and word bigram for feature extraction in SVM and Naive Bayes models.

The second approach, we conduct experiments using single model with deep neural network for multi-label classification. We implement CNN model architecture from [7], but we use sigmoid activation function for computing the output. This architecture consists of 1D convolutional layer, max pooling layer, and hidden dense layer. We also experiment on a different architecture by connecting, a simplified version of LSTM, Gated Recurrent Unit (GRU) to the CNN layer. GRU helps improving performance especially for smaller datasets [13].

We use pre-trained FastText model, which had been performed on our domain-specific dataset, to extract input vectors. FastText model was trained for 200 dimensional feature vectors with 5 windows. The sentence matrix, consists of the concatenation on word vectors, is sent into the CNN layer for the first architecture. Meanwhile, in the second architecture, each word vector runs into the GRU which wrapped in Bidirectional layer.

CNN layer can recognize patterns within sentences by utilizing the kernel size. This kernel size works like word ngrams, where a convolution filter with kernel size 2 is similar to bigrams so it tracks 2 word vectors for each computation to the end of matrix. The output of CNN layer is transferred into a max pooling layer to find the maximum value for each filter. These processes result in losing the information of word position. GRU can capture the sequential information, since it has an architecture of RNN. GRU provides update gate and reset gate to determine which information is kept or removed. The final layer, a dense layer, contains 13 hidden nodes which are the total aspect categories in our dataset. This layer computes the output, for predicting aspects, using sigmoid activation function.

$$\text{Hamming Score} = \frac{1}{n} \sum_{i=1}^n \frac{|\hat{Y}_i \cap Y_i|}{|\hat{Y}_i \cup Y_i|} \quad (1)$$

We used two different metrics to measure the performances which are example-based and label-based [14]. Example-based is used to compare how close the result of predicted labels to the actual labels, while label-based measures the performance of models on identifying each label separately.

⁵<https://pypi.org/project/Sastrawi/>

We employed Hamming Score to calculate the accuracy of multi-label models and we analyzed each label with F1-score. The equation of Hamming Score is the fraction between the intersection of predicted ($\hat{Y}_i$) and actual (Y_i) values with the union between them.

IV. EXPERIMENTS AND ANALYSIS

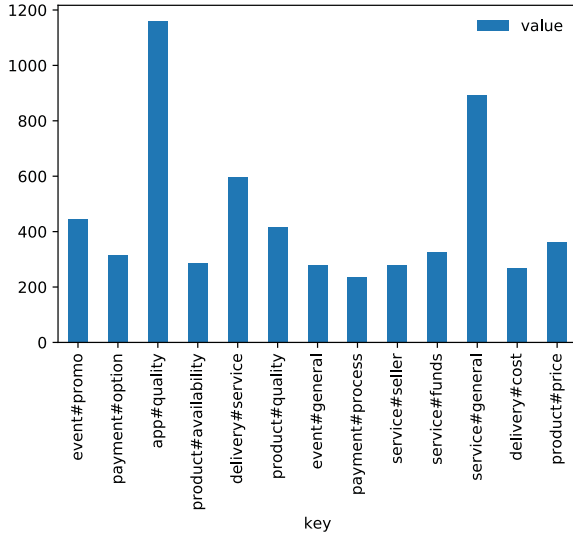


Fig. 2. Label distribution

On performing the experiments, we first collected the dataset from Google Play Store and Apple App Store using google-play-scraper⁶ and app-store-scraper⁷ respectively. We obtained 64113 unduplicated reviews from 6 e-commerce applications in Indonesia. Then, we randomly selected and annotated 3748 reviews. The aspects distribution is shown in Fig. 2. Aspect *app#quality* dominates others in total count. It corresponds to the sources of datasets which are from mobile application marketplace reviews. The *event#promo* aspect is also among the highest count in the distribution, indicating that users are interested to this aspect. On the other hand, the overall graph represents a skew distribution, means it suffers from imbalanced dataset problem. Furthermore, we also analyzed the correlation between aspects and they fell within the range of -0.18 to 0.13. It showed that the aspects were independent from each other, because the correlation values are close to 0.

The experiments for SVM and Naive Bayes models were conducted using scikit-multilearn⁸ library. The dataset is split into 70% training and 30% testing sets. The aspects are transformed into multi-label binary. We used 5 folds cross-validation to run these models. Gaussian Naive Bayes is employed for Naive Bayes model, while we executed two

scenarios for SVM model, i.e. without and with class weights (SVM+Weighting).

The next experiment, for single model, we implemented neural network with Keras⁹ library. We used 64 filters for kernel size 2 and ReLU activation function in convolutional layer. We set 128 units (output dimension) for GRU. We also added dropout 0.5 before proceeding to the convolutional layer. Finally, the model was trained using Adam optimizer with 0.001 learning rate and binary cross entropy for calculating the loss function. We assigned the value for the batch size to 32 and added callback functions to save the best model and insert early stopping operation to prevent overfitting. For calculating the metrics, we set the threshold by 0.4.

TABLE II
ONE-VS-ALL APPROACH ACCURACY

Model	Hamming score
Gaussian NB	0.318
SVM	0.637
SVM+Weighting	0.675

First, we measured the performance using example-based metric to analyze the accuracy of our models. In one-vs-all experiment, we found that Gaussian NB suffered the most in an imbalanced dataset, as it only scored 0.32 accuracy. Increasing as much as twice the result of Gaussian NB, SVM model achieved 0.64 accuracy and got better when we added the class-weights. SVM+Weighting earned the best result thus far with 0.67 accuracy. The addition of class weights shows positive improvement for classifying our imbalanced dataset.

TABLE III
NEURAL NETWORK APPROACH ACCURACY

Model	Hamming score
CNN	0.653
GRU+CNN	0.713

For the next experiment, neural network single model approaches achieved a decent result although the dataset is small and unbalanced. CNN model got 0.65 accuracy which is close to reach SVM+Weighting result. On the other hand, GRU+CNN increased 6% in accuracy from the CNN model and successfully beat the best result in one-vs-all approach by 0.71 accuracy. The result on the architecture of GRU+CNN shows that the context in text review is also important to classify the aspects.

We then chose two best models for analysing on each aspect's identification using label-based metric F1-score¹⁰. GRU+CNN won over common approach in multi-label task, one-vs-all SVM+Weighting model, almost in all aspects except for *payment#option* and *product#availability*. Meanwhile,

⁶<https://github.com/facundooolano/google-play-scraper>

⁷<https://github.com/facundooolano/app-store-scraper>

⁸<http://scikit.ml/>

⁹<https://keras.io/>

¹⁰https://scikit-learn.org/stable/modules/generated/sklearn.metrics.classification_report.html

TABLE IV
LABEL-BASED F1-SCORE RESULT

Aspect	SVM+Weighting	GRU+CNN
app#quality	0.76	0.79
delivery#cost	0.87	0.87
delivery#service	0.78	0.82
event#promo	0.85	0.88
event#general	0.72	0.83
payment#option	0.88	0.86
payment#process	0.57	0.65
product#availability	0.61	0.60
product#price	0.85	0.90
product#quality	0.73	0.73
service#funds	0.86	0.86
service#general	0.62	0.65
service#seller	0.63	0.64

three aspects, including *delivery#cost*, *product#quality*, and *service#funds* resulted in a tie. Overall, GRU+CNN model showed a promising result for both example-based and label-based performances.

In GRU+CNN model result, 7 aspects out of 13 aspects obtained a good F1-score, which greater than 0.80. On the contrary, *payment#process*, *product#availability*, *service#general*, and *service#seller* only achieved around 0.60 F1-score. Due to the amount of *payment#process* aspect is the least in our dataset, the model had difficulty to identify this aspect especially when the term 'COD' (*Cash On Delivery*) was involved. Term 'COD' usually refers to *payment#option* but in some cases, users use the term to explain a payment process of COD. For example, "*suka sama tampilan nya rapih, pengiriman cepat dan pembayaran ga repot apa lagi buat yang cod cuman tolong produk2nya dibanyakin dan dilengkapi lagi ya*" ("*I like the clean interface, fast delivery, and the payment is not difficult especially cod, but please add more products variant*"). The model identified the *payment#option* in the text instead of *payment#process*.

For *product#availability*, the model failed to identify the aspect when the reviews specifically mentioned the product name. For example, "*Tiket Persib nya juga udah ga ada . Mohon bantuannya minta keadilannya. Mohon bantu*", ("*Persib ticket is not available. Please help.*"), the "*Persib ticket*", which is a product's name, was not identified by the model as *product#availability*. This problem might be solved by adding part-of-speech tag (POS tag) feature or Named-entity recognition (NER) for specifically recognized product's name in the future experiments.

While the model was able to identify *service#general* in 'customer services' related reviews, it still found difficulty to recognize *service#general* where users implicitly complained about policies. In "*biasanya dapat untung dari berat, sekarang uang yang lebih dari berat itu diambil . :(dulu mah kita dapat sedikit untung dari berat volume*" ("*I used to get slightly extra profit from item's weight, but now we also could not*

take that money"), the user talked about profit sharing policy which supposedly fell into *service#general* aspect, but the model failed to identify it. On the contrary, the model often mistook *service#general* for *service#seller* when the reviews talked about communication with seller. Moreover the model also missed on wrong-delivered items complains. It usually was directed towards the seller's fault but the model identified it as *delivery#service* aspect.

V. CONCLUSION

In this paper, we analyzed e-commerce reviews from different source of dataset. We found that some important aspects had not been covered in previous studies, such as *promo* and *payment*. Since many people access e-commerce through mobile application, our dataset can assess the e-commerce platform's aspects more thoroughly and more varied.

In aspect detection model, we compare one-vs-all approach to single model approach. Although the aspects in dataset has no dependencies to each other, which is the advantage of one-vs-all, the GRU+CNN single model successfully achieve the best result in our experiment with 71.3% accuracy. The same model also generally gave a good performance in identifying each aspect separately, as more than a half of aspects obtained greater than 80% F1-score. However, the rest of aspects still need to be analyzed further to improve the individual aspect identification. For future experiments, we can collect more variety of data, define the aspects in more detail explanation to reduce the ambiguity, and enrich the model with additional features such as POS tag or NER. Besides, collecting more data will potentially give advantages to the neural network models, since they are known to perform better in a larger dataset.

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